Exploring the world of weakly-supervised segmentation

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1 Introduction

Semantic Segmentation operates by classifying the label for each pixel in the image, where the pixels are the segments in the image [14]. Furthermore, the label is picked from the label set, which is already defined [14]. Semantic Segmentation is important because it helps computer machines understand images at a detailed level.

Fully supervised segmentation is a learning method where a model is trained with images at the pixel level where each pixel has a precise label, which is known as the ground truth [11]. The challenges of fully-supervised segmentation are that finding images that have labels for each pixel is difficult espcially in the specific contexts, and it is time-consuming to annotate every pixel for every image [11].

The weak supervision challenge is that there are insufficient fully annotated datasets available and there are some datasets where parts of the datasets are labeled in terms of pixels. So the model training is hard and as a result the model does not know how to localize an item in an image. To overcome this, there is another learning method, weakly-supervised segmentation , where the model can train on less detailed pixel-level annotation [4]. This method is important as it overcomes the problem of the time taken on annotating images to a high level. Furthermore, in some contexts detailed annotation is harder to scale [4]. Therefore using weakly-supervised segmentaion helps reduce the cost of annotation.

Commonly used weakly supervised semantic segmentation methods are Class Activation Maps (CAMs), Multiple Instance Learning (MIL), Self-Training and Conditional Random Field. CAMs aim to find where the item is in an image by using the Convolutional Neural Network (CNN) which is used for classification training [8]. Then, in the last layer of the CNN, the feature maps acquire the general layout of the image. After this, the fully connected layers add weights to the feature maps [8]. Then CAM is generated for each class by the feature maps and this multiplied by the weights and totaled [8]. This result conveys which areas of the heatmap aided the model for choosing the classification class [8]. MIL also examines the images by annotating the group of images but does not label every individual image [3]. For example, if there are mutiple images, with one containing a dog, the whole group of images is considered as dogs but it does

not know which image contains the dog [3]. Self training is performed in a loop where it learns the prediction output from itself and increases its understanding using CAMs [13]. CRFs aid with pixel labelling and enhances image segmentation mask [9]. Using the CAM map, CRF looks at the pixel colours, the other pixels nearby, how close the pixels are and labels the pixels [9].

The MRP points that we will address are, the supervision problem will be attempted to solve using Grad-CAM as it is a commonly used weakly-supervised segmentation method. Then we will use our implemented weakly-supervised segmentation framework baseline version to compare with a fully supervised model. The selected open ended question (OEQ) to address, "How do different combinations of confidence thresholding and class-adaptive noise injection during pseudo-mask generation affect the overall mean IoU and per-class segmentation robustness in a weakly-supervised segmentation setting?". The selected hypothesis is "Tailoring pseudo-mask refinement and noise injection parameters to individual classes will improve both (a) overall mIoU and (b) the worst-performing class's IoU, by accounting for class-dependent mask quality and imbalance". The OEQ question will be addressed via the experiments conducted below. We are using the experiments with the MRP points, on the weakly-supervised segmentation framework baseline to try boost the metric results. In the end, we produced the final weakly-supervised segmentation framework which is the code from baseline but with the code additions from experiments that performed the best in terms of the metrics.

Weakly-supervised segmentation is useful to evaluate on the Oxford-III Pet Dataset as the dataset has 37 types of pet breeds where each class has 200 images. There are approximately 7400 images in this dataset and each image has the ground truth containing the breed, head region of interest which is the pet's head location in the image which is tracked via bounding box and pixel level trimap segmentation [7].

2 Methods

The methodology consists of three parts where in the first part, we created a fully supervised model to compare with the weakly-supervised segmentation framework baseline where both use the fully convolutional ResNet 50 network. In the second part, we experimented with eight different experiments on the weakly-supervised segmentation framework baseline to try and boost the metric results. In the final part, we produced the final weakly-supervised segmentation framework improved version.

The weakly-supervised segmentation framework baseline was implemented by loading the Oxford-III Pet Dataset with the masks. Then, this dataset was split where 80% was used for training and 20% was used for internal validation. The ResNet-50 Classifer was trained using image level pet breeds. CRF is used on the Grad-CAM heatmaps to make the pseudo-mask more accurate and high quality. The pseudo-masks were produced with Grad-CAM where heatmaps are turned into rough foreground masks. The fully connected network ResNet-50 Segmenter

was trained on generated pseudo-masks in the previous step and chooses the best IoU from the checkpoint. In the evaluation stage, we calculate the IoU, Dice and pixel accuracy on the true masks and store the sample predictions.

Studying the OEQ question, the addition of category-specfic enhancement of pseudo-mask quality is appropriate. Our aim was to examine the confidence threshold and label noise ratio as the confidence threshold controls the amount of the Grad-CAM that is preserved. This is important as if most of the Grad-CAM is not kept, this will affect the pseudo-mask generation meaning it will also affect the training on the segmentation model. The label noise ratio retains the percentage of mask pixels that are inverted, to create situations with flaws so we can review how robust the model is.

The methodology for the OEQ question was to use part of the dataset and perform a small grid search with confidence values of 20% ,40% ,60% and 80% with different synthetic noise ratio levels of 0%, 10% , 20% and 30%. In the code implementation, we needed to alter the "percentile_threshold function" as this will execute one of the confidence threshold values specified above on the Grad-CAM result. The alteration of the "flip_mask_pixels" function is needed as this was infusing noise to the mask. The end goal was to review which pair of values give us the best IoU for each category.

3 Experiments

A comparison of weakly-supervised segmentation framework baseline with the fully supervised model was performed where both use similar architecture to make the comparison fair in this section.

There are 8 experiments, experiment 1 was to find a way to prevent overfitting so we can enhance the validation accuracy and this was done by adding dropout layers for both classfier and segmentation models. These dropout layers were only added between Bottleneck blocks [1]. The dropout numbers were adjusted by scaling down so the in-depth layers do not get altered by the dropout in comparisons to the not so deep layers. Addtionally hyperparameter exploration was performed with the dropout levels.

Experiment 2 was to refine Grad-CAM and pseudo-mask generation. This was attempted to improve in two ways, one was Adversarial Erasing from Yunchao's paper and this works by erasing the mass powerful zone in the image [12]. This pressured the Grad-CAM to view other areas. The second technique is Consistency Regularisation and this utilized a variety of images which are transformed and is used for self-supervision in network learning [10].

Experiment 3 was to experiment with the CRF parameters and review if the metrics improve. This was experimented using Optuna which is a hyperparameter optmization frameworks. This framework explores the set of CRF parameters to find the leading CRF parameters using the Bayesian Optimisation method which produces the highest IoU for the segmentation model [6]. The CRF parameters which are explored by Optuna are compat_gauss, srgb, sxy_gauss, sxy_bilat and n_iters.

Experiment 4 was to examine metric results when the segmentation model changed from FCN-ResNet50 to DeepLabV3-ResNet50. The reason why we swapped the model to DeepLabV3-ResNet50 is because this model acquires multi-scale contextual information which is a stronger model compared to FCN-ResNet50 as this is considered as a lightweight model. Further, the AdamW scheduler was enforced to increase the chances of improving accuracy [5]. The AdamW scheduler operates by altering the learning rate which is adaptive and weight decay regularisation. This aids with the studying of noisy supervision data as the loss is guided.

The fifth experiment was to fine tune the segmentation model with 10% ground truth after the segmentation training. The finetuning was performed using a specfically altered dataset where there was an even number of classes which were acquired to make sure the experiment is fair and there is no class that is higher than another to prevent overlearning. For the fine tuning, the learning rate value was small to ensure the model keeps the learned parameters in its memory.

Experiment 6 is to experiment with pseudo-masks by adding synthetic noise and comparing the segmentation IoU metric with the baseline and improved version. This experiment aids with answering the OEQ question.

The seventh experiment was to attempt to pre train encoder with Bootstrap Your Own Latent (BYOL) pet images with no labels and then compare with the weakly-supervised segementation framework baseline using the IoU metric. BYOL works by transforming the same image in different ways and trains one network on one image with one view [2]. Then another network trains on the same image with a different view. The first network tries to gain the same understanding as the second network even though they both use different views on the same image [2].

The final experiment was to compare the weakly-supervised segmentation framework baseline with bounding boxes created via CAM contours to help with the pseudo-mask generation as this deeply focuses on the mask before the segmentation model training.

4 Results

When we the ran the weakly-supervised segementation baseline and the fully supervised model, the fully supervised model scored an IoU of 0.9197, dice score of 0.8261 and pixel accuracy of 0.9287 whereas the weakly-supervised segementation baseline model scored lower values on all three metrics (refer to Table 2). Next we performed the experiments and tried to improve the weakly-supervised segementation baseline model.

Experiment 1 was not experimented fully due to computation resources and time as each epoch took around 1 hour to 1 hour and 30 minutes. During the code execution, the computer would sometimes crash too, so there are some partial results. The results we obtained are for dropout value 0.1, the loss was around 3.77-3.68. The training accuracy ranged from 3.02% - 3.16% and the validation

Experiment	IoU	Dice Score	Pixel Accuracy
1	-	-	-
2	0.1042	0.2047	0.2083
3	0.1723	0.2523	0.2232
4	0.0422	0.1472	0.1566
5	0.2574	0.2980	0.3201
6	0.2817	0.3094	0.3379
7	0.0389	0.1449	0.1555
8	0.0102	0.1185	0.1316

Table 1. Experiment Results

Configuration Setup	IoU	Dice Score	Pixel Accuracy
Weakly-supervised segmentation baseline	0.0541	0.1586	0.1662
Weakly-supervised segmentation improved	0.2887	0.2968	0.334
Fully supervised	0.9197	0.8261	0.9297

Table 2. Comparison of Configurations

accuracy ranged from 2.17% to 1.77%. For dropout value 0.2, the loss was around 4.01-3.80. The training accuracy ranged from 2.92% to 2.51% and the validation accuracy was 2.17%. Reviewing these results, dropout value helped the training loss more as there was a larger discrepancy in the training loss. However, the validation accuracy did not change at all.

Experiment 2 using the Adversarial Erasing technique, the pseudo-mask had more region coverage after erasing the powerful region conveying that this technique aided the Grad-CAM (refer to Fig 1 appendix). Additionally, the consistency regularisation improved the region coverage compared to the weakly-supervised framework baseline implementation (refer to Fig 2 appendix).

Experiment 3 results helped us find the leading CRF parameters. The compat_gauss is 4, sxy_gauss is 2.5974, compat_bilat is 1, n_iters is 8 and sxy_bilat is 30.8892. The specified CRF parameters updated the segmentation masks and gave us an IoU result of 0.1736. The usage of Optuna assisted with the quality of segmentation masks using spatial smoothing and colour similarity.

Experiment 4 did not perform well with DeepLabV3-Resnet50 model. The results produced were: the top IoU was 0.0422, the dice score is 0.1472 and the pixel accuracy scored 0.1566. The pseudo-masks produced by Grad-CAM can localize the item in the image well; however, there was difficulty of finding the borders of the item in the image. In contrast to this, the training loss dropped as each epoch passed, conveying that the model is learning but it is not enough for the IoU to be stablized.

Experiment 5's results gained were terrible where the model could not gain any memory on the learned parameters. So the pseudo-masks did not look great. After exploring why this happened, we identified the masks only acquired some features of the animals instead of covering the whole animal body. To overcome this, we reviewed the foreground coverage and discovered 90% of the images,

the foreground covered a large amount (>= 80%) of the area in the image. This means the threshold needs to be increased in the mask production which potentially could produce better pseudo-masks. After increasing the threshold, the mask coverage covered more parts of the animal body however the IoU metric reduced by 1-2%. The final prediction image produced conveyed that the mask object edges are more defined and can see the outline of the animal.

In experiment 6, out of the threshold and noise ratio combinations (refer to noise results in appendix), the threshold value of 40% and the noise ratio value of 30% produced the best result. This addition to the weakly-supervised segmentation improved version allowed us to receive an IoU of 0.2887, dice score of 0.2968 and pixel accuracy of 0.3340.

Experiment 7 where comparing the use of BYOL, when BYOL was not used, the pseudo-masks did not display the item location accurately. Furthermore, the pet body shape was not clearly conveyed. However, the GradCAM heatmap selected areas where the pet is and areas that are local to the pet. With the usage of BYOL, the GradCAM heatmap is spread out and the prediction from the segmentation model is not strong and does not capture the pet location at all. This could be because there is not enough fine tuning after the application of BYOL.

In experiment 8, the usage of box guided segmentation did not produce higher scores than image level (baseline) as its IoU, dice score, accuracy scores produced were 0.0102, 0.1185 and 0.1316. However in image level approach , the scores were already higher (0.0541,0.1586,0.1662). Overall, it seems that the image level approach had more flexibility in generating the masks by excluding the usage of box constraints.

5 Discussion

Summarising our key findings, some experiments performed well, however the problem is merging two or more experiments' methodologies that perform well and trying to improve the result. As there was insufficient time to do this as each execution of the code took around 8 hours and there was a risk of deteriorating the results. The combination of some experiments should be done in the future but not all as some may collide with their methodology. Even if there is a risk of getting low results so we can get further understanding on why this is happening.

6 Conclusion

We managed to improve the weakly-supervised segmentation framework baseline via confidence tuning and noise injection which produced 0.2887 IoU (improved) compared to 0.0541 IoU (baseline) which is an improvement but not as substantial as the fully supervised model.

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A Appendix

Fig. 1. Adversarial Erasing

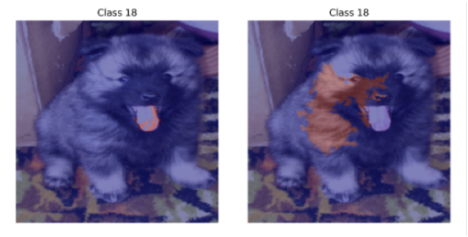

Fig. 2. Consistency Regularization Technique

Noise Level	Best Val IoU
0% (none)	0.2741
10%	0.2718
20%	0.2699
30%	0.2755

Table 3. Noise level

Fig. 3. Fully Supervised Result

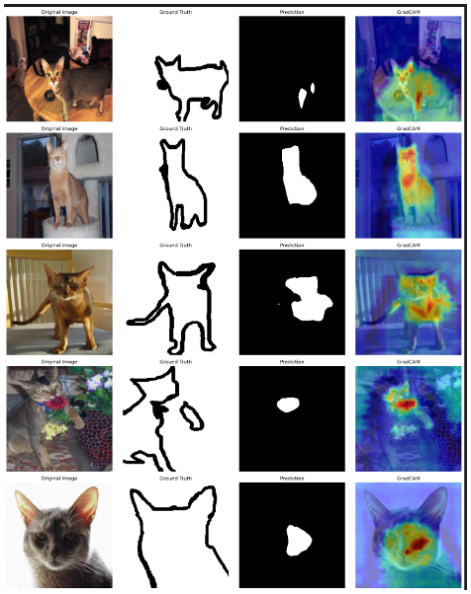

Fig. 4. Weakly-supervised Segmentation Baseline Result

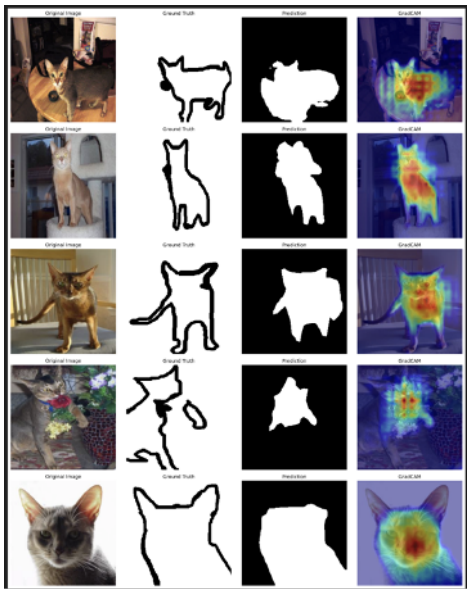

Fig. 5. Weakly-supervised Segmentation Improved Result